

in the fact that there's no human input whatsoever at any stage of the "creative" process. Once AARON starts drawing, a human can't influence it in any way. In fact, the only thing that can influence it is AARON itself.

Of course, there has to be human input at some point for AARON to "see" the outside world. Over time, Cohen added several models along a three-dimensional axis to AARON, thus making it more aware of the visual world. In fact, he stated that in one instance when Cohen was developing the knowledge of structure of heads and faces to make AARON draw a facsimile of a friend, the AI started generating make-believe people that looked eerily like some of the people Cohen knew.

AARON has the unique ability to work on feedback. What this means is that each single line that AARON draws is registered by the machine, and it goes through a series of thousands of decisions to decide what to draw next. Of course, given modern mechanics, this happens quite a lot faster than it sounds.

But there's still a fundamental difference between the way AARON draws and the way a human draws. For AARON, any painting starts with the foreground object and works its way backwards to the background. Given the feedback system it works on, the "whole" picture isn't seen from the start like a human sees it. Indeed, human artists have a "vision" that they try to replicate on canvas, whereas AARON uses a systematic, step-by-step approach to reaching a full picture at the end, even though it might not know what that picture is going to be at any point of the creation process.

Kurzweil and Cohen have also worked on other programs about the intersection of creativity, art and computers, notably a robot that generates poetry. Similarly, composer and computer scientist David Cope's EMMY could write beautiful symphonies reminiscent of the works of Bach and Mozart.

Can AI have a sense of humour?

Getting back to the core conversation about creativity, it's interesting to know that its usage in the human brain is not just in creating something new, but also in understanding the link between two seemingly disjointed artefacts. A sense of humour is perhaps the greatest example of this.

The human ability to have a sense of humour is dependent on understanding how one object is linked with another. As Margaret Bolden explains: